



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Software Construction And Development lab Report - 13

NAME: M Abdul Rafay (FL23791)

PROGRAM: BSSE (5th Semester)

COURSE: Software Construction and Development

SUBMITTED TO: Sir Ahsan Arif

DATE: 9/21/2025

DAY: Sunday

Lab Task

1. Setup Initial Test Structure

Create a basic test structure for a simple function or class. Write a test that verifies a basic functionality (e.g., a function that adds two numbers).

2. Write a Failing Test

Write a test that initially fails because the code under test is not yet implemented. Write a test case that expects a certain behaviour from the code (e.g., expecting a function to return a specific result).

3. Implement Minimum Code to Pass Test

Implement the minimum code necessary to make the failing test pass. Write the code to fulfill the requirements of the test case, ensuring it passes the test without adding unnecessary complexity.

4. Refactor Code

Refactor the code to improve design, readability, and maintainability. Review the implemented code and refactor it while ensuring all tests continue to pass. Remove duplication, improve naming, and apply best practices.

Code :

```
public class Password {

    private String password;

    public Password(String password) {
        this.password = password;
    }

    // Updated password validation rules
    public boolean isValid() {
        return isLengthValid() &&
```

```

        hasUpperCase() &&
        hasLowerCase() &&
        hasDigit() &&
        noSpaces();
    }

    private boolean isValidLength() {
        return password.length() >= 5 && password.length() <= 10;
    }

    private boolean hasUpperCase() {
        return password.matches(".*[A-Z].*");
    }

    private boolean hasLowerCase() {
        return password.matches(".*[a-z].*");
    }

    private boolean hasDigit() {
        return password.matches(".*[0-9].*");
    }

    private boolean noSpaces() {
        return !password.contains(" ");
    }

    // Main function to manually test cases
    public static void main(String[] args) {

        // Test Case 1: Too short
        Password p1 = new Password("Ab1");

```

```
System.out.println("TC1 (Ab1) → " + p1.isValid());
```

```
// Test Case 2: Missing uppercase
```

```
Password p2 = new Password("hello1");
```

```
System.out.println("TC2 (hello1) → " + p2.isValid());
```

```
// Test Case 3: Missing digit
```

```
Password p3 = new Password("Hello");
```

```
System.out.println("TC3 (Hello) → " + p3.isValid());
```

```
// Test Case 4: Valid password
```

```
Password p4 = new Password("Hello1");
```

```
System.out.println("TC4 (Hello1) → " + p4.isValid());
```

```
// Test Case 5: Contains space
```

```
Password p5 = new Password("He llo1");
```

```
System.out.println("TC5 (He llo1) → " + p5.isValid());
```

```
// Test Case 6: Too long
```

```
Password p6 = new Password("Hello12345678");
```

```
System.out.println("TC6 (Hello12345678) → " + p6.isValid());
```

```
// Test Case 7: Missing lowercase
```

```
Password p7 = new Password("HELLO1");
```

```
System.out.println("TC7 (HELLO1) → " + p7.isValid());
```

```
// Test Case 8: Perfect valid password
```

```
Password p8 = new Password("Abc12");
```

```
System.out.println("TC8 (Abc12) → " + p8.isValid());
```

```
}
```

```
}
```

Output:

```
Output - lab8 (run) X
run:
TC1 (Ab1) ? false
TC2 (hello1) ? false
TC3 (Hello) ? false
TC4 (Hello1) ? true
TC5 (He llol) ? false
TC6 (Hello12345678) ? false
TC7 (HELLO1) ? false
TC8 (Abcl2) ? true
BUILD SUCCESSFUL (total time: 0 seconds)
```